# A special case of the Gamma-distribution

## Parametrisation

We consider this distribution

$$\pi(y) = \frac{1}{\Gamma(a)} y^{a-1} \exp(-y), \qquad a > 0, \quad y > 0,$$

where $\mathrm{E}(y) = \mu = a$.

## Link-function

The linear predictor $\eta$ is linked to the mean $\mu$ using a default log-link

$$\mu = \exp(\eta)$$

## Hyperparameter

None.

## Specification

- `family="gammajw"` for regression models and `family="gammajw.surv"` for survival models.
- Required arguments: for `gammajw.surv`, $y$ (to be given in a format by using `inla.surv()`), and for `gammajw`, $y$.

### Hyperparameter spesification and default values

**gammajw:**

**doc** `A special case of the Gamma likelihood`

**hyper**

**survival** `FALSE`

**discrete** `FALSE`

**link** `default log neglog`

**pdf** `gammajw`

**gammajwsurv:**

**doc** `A special case of the Gamma likelihood (survival)`

**hyper**

**theta1**

**hyperid** `58200`
**name** `beta1`
**short.name** `beta1`
**output.name** `beta1 for GammaJW-Cure`
**output.name.intern** `beta1 for GammaJW-Cure`
**initial** `-7`

**fixed** FALSE
**prior** normal
**param** -4 100
**to.theta** function(x) x
**from.theta** function(x) x

**theta2**

**hyperid** 58201
**name** beta2
**short.name** beta2
**output.name** beta1 for GammaJW-Cure
**output.name.intern** beta1 for GammaJW-Cure
**initial** 0
**fixed** FALSE
**prior** normal
**param** 0 100
**to.theta** function(x) x
**from.theta** function(x) x

**theta3**

**hyperid** 58202
**name** beta3
**short.name** beta3
**output.name** beta3 for GammaJW-Cure
**output.name.intern** beta3 for GammaJW-Cure
**initial** 0
**fixed** FALSE
**prior** normal
**param** 0 100
**to.theta** function(x) x
**from.theta** function(x) x

**theta4**

**hyperid** 58203
**name** beta4
**short.name** beta4
**output.name** beta4 for GammaJW-Cure
**output.name.intern** beta4 for GammaJW-Cure
**initial** 0
**fixed** FALSE
**prior** normal
**param** 0 100
**to.theta** function(x) x
**from.theta** function(x) x

**theta5**

**hyperid** 58204
**name** beta5

**short.name** beta5
**output.name** beta5 for GammaJW-Cure
**output.name.intern** beta5 for GammaJW-Cure
**initial** 0
**fixed** FALSE
**prior** normal
**param** 0 100
**to.theta** function(x) x
**from.theta** function(x) x

**theta6**

**hyperid** 58205
**name** beta6
**short.name** beta6
**output.name** beta6 for GammaJW-Cure
**output.name.intern** beta6 for GammaJW-Cure
**initial** 0
**fixed** FALSE
**prior** normal
**param** 0 100
**to.theta** function(x) x
**from.theta** function(x) x

**theta7**

**hyperid** 58206
**name** beta7
**short.name** beta7
**output.name** beta7 for GammaJW-Cure
**output.name.intern** beta7 for GammaJW-Cure
**initial** 0
**fixed** FALSE
**prior** normal
**param** 0 100
**to.theta** function(x) x
**from.theta** function(x) x

**theta8**

**hyperid** 58207
**name** beta8
**short.name** beta8
**output.name** beta8 for GammaJW-Cure
**output.name.intern** beta8 for GammaJW-Cure
**initial** 0
**fixed** FALSE
**prior** normal
**param** 0 100
**to.theta** function(x) x

**from.theta** `function(x) x`

**theta9**

**hyperid** `58208`
**name** `beta9`
**short.name** `beta9`
**output.name** `beta9 for GammaJW-Cure`
**output.name.intern** `beta9 for GammaJW-Cure`
**initial** `0`
**fixed** `FALSE`
**prior** `normal`
**param** `0 100`
**to.theta** `function(x) x`
**from.theta** `function(x) x`

**theta10**

**hyperid** `58209`
**name** `beta10`
**short.name** `beta10`
**output.name** `beta10 for GammaJW-Cure`
**output.name.intern** `beta10 for GammaJW-Cure`
**initial** `0`
**fixed** `FALSE`
**prior** `normal`
**param** `0 100`
**to.theta** `function(x) x`
**from.theta** `function(x) x`

**survival** `TRUE`

**discrete** `FALSE`

**link** `default log`

**pdf** `gammajw`

## Example

In the following example we estimate the parameters in a simulated example.

```
n <- 300
x <- rnorm(n, sd = 0.3)
eta <- 1 + x
mu <- exp(eta)
y <- rgamma(n, shape = mu, scale = 1)
r <- inla(y ~ 1 + x,
          data = data.frame(y, x),
          family = "gammajw",
          control.compute = list(cpo = TRUE),
          control.fixed = list(prec.intercept = 0.01),
          verbose = TRUE)
```

```
summary(r)

yy <- inla.surv(y, event = 1)
rr <- inla(yy ~ 1 + x,
           data = list(yy = yy, x = x),
           family = "gammajwsurv",
           control.compute = list(cpo = TRUE),
           control.fixed = list(prec.intercept = 0.01),
           verbose = TRUE)
summary(rr)

print(r$summary.fixed - rr$summary.fixed)
```

## Notes

None.